

Step-by-Step Elastic Kubernetes Service (EKS) Cluster Setup with Auto Mode via AWS Console and eksctl



By

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What is EKS Auto Mode?

EKS Auto Mode is a management enhancement for Amazon Elastic Kubernetes Service (EKS) that simplifies the operational management of Kubernetes clusters and extends AWS's control beyond the cluster itself. It automates the setup and management of the infrastructure required for running workloads, including compute, networking, storage, and scaling, while integrating deeply with AWS services like EC2, EBS, and ELB.

Key Features of EKS Auto Mode

1. **Streamlined Management:** Production-ready clusters with minimal operational overhead and automation of key tasks.
2. **Automated Compute and Scaling:**
 - Automated node lifecycle, scaling with Karpenter, and GPU support.
 - Regular node replacements for security (21-day max lifetime).
3. **Networking and Load Balancing:**
 - Automates Pod/service networking with IPv4/IPv6 support.
 - Integrates with AWS Elastic Load Balancers for seamless traffic management.
4. **Cost Optimization:**
 - Automatically consolidates workloads and terminates unused nodes.
 - Supports cost-saving configurations like Spot instances.
5. **Security:**
 - Immutable AMIs, SELinux enforcement, and no direct SSH/SSM access.
 - Regular updates to enhance security posture.
6. **Customizability:**
 - Custom NodePools and NodeClasses for specialized workloads.
 - Support for DaemonSets to add custom monitoring/services.
7. **AWS Integration:** Fully integrates with EC2, EBS, ELB, and other AWS services.

EKS Auto Mode Pricing:

EKS Auto Mode Pricing simplifies Amazon EKS cluster management by automatically selecting and scaling the underlying infrastructure. Pricing includes:

1. **EC2 Instance Costs:** Standard charges for the EC2 instances provisioned for your workloads.

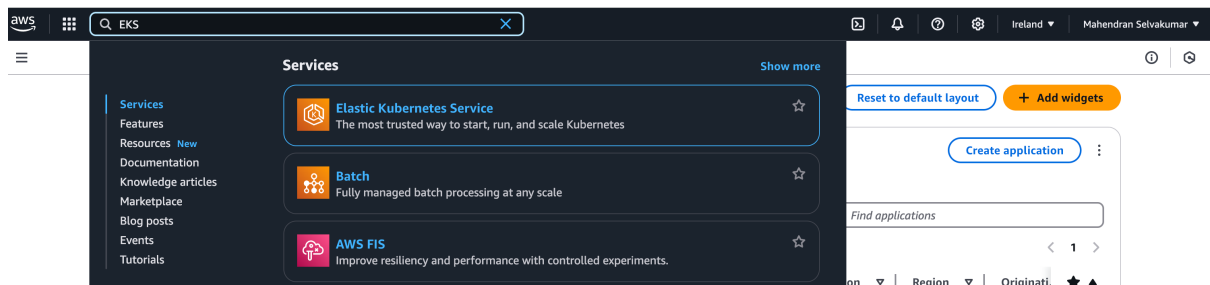


2. **Management Fee:** A per-hour fee for each EC2 instance, which varies by instance type, to cover the automation of infrastructure management.

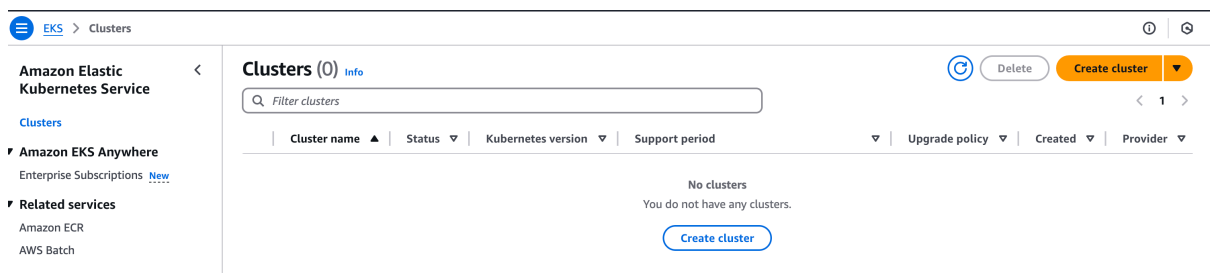
Auto Mode dynamically optimizes resource usage and scales across Availability Zones, ensuring high availability and cost-effectiveness. You only pay for the compute and management fees associated with your active workloads

Step 1: Create EKS Cluster with Auto Mode

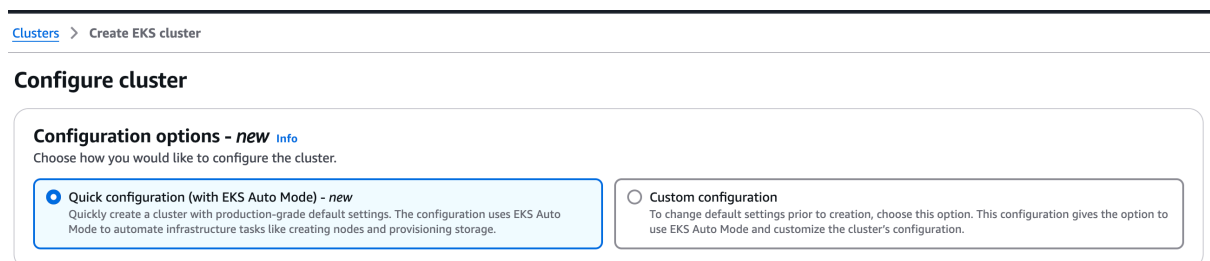
Open the AWS Management Console, search for **EKS**, and click on **Elastic Kubernetes Service**



Click on **Create cluster**



Select **Quick configuration (with EKS Auto Mode)** to enable **EKS Auto Mode**



Enter the **EKS cluster name** and select the **Kubernetes version**



Cluster configuration

Name

Use the auto-generated name or enter a unique name for this cluster. This property cannot be changed after the cluster is created.

mahi-eks-auto-mode



The cluster name should begin with letter or digit and can have any of the following characters: the set of Unicode letters, digits, hyphens and underscores. Maximum length of 100.

Kubernetes version

Select Kubernetes version for this cluster.

1.31



Click **Create recommended role** to create the **Cluster IAM role**

Cluster IAM role

Select the Cluster IAM role to allow the Kubernetes control plane to manage AWS resources on your behalf. This cannot be changed after the cluster is created. To create a new custom role, follow the instructions in the [Amazon EKS User Guide](#).

Select role



Create recommended role

Node IAM role

Nodes need an EC2 Instance IAM Role to launch and register with a cluster. To create a new custom role, follow the instructions in the [Amazon EKS User Guide](#).

Select role



Create recommended role

Select **AWS Service** as the trusted entity

IAM > Roles > Create role

- Step 1
- Select trusted entity**
- Step 2
- Add permissions
- Step 3
- Name, review, and create

Select trusted entity

Trusted entity type

☒ AWS service

Allow AWS services like EC2, Lambda, or others to perform actions in this account.

☐ AWS account

Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.

☐ Web identity

Allows users federated by the specified external web identity provider to assume this role to perform actions in this account.

☐ SAML 2.0 federation

Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.

☐ Custom trust policy

Create a custom trust policy to enable others to perform actions in this account.

Select **EKS** as the service and **EKS-Auto Cluster** as the use case, then click **Next**



Use case

Allow an AWS service like EC2, Lambda, or others to perform actions in this account.

Service or use case

EKS

Choose a use case for the specified service.

Use case

- ☐ EKS - Service
Allows EKS to manage clusters on your behalf.
- ☐ EKS - Cluster
Allows the cluster Kubernetes control plane to manage AWS resources on your behalf.
- ☐ EKS - Nodegroup
Allow EKS to manage nodegroups on your behalf.
- ☐ EKS - Fargate pod
Allows access to other AWS service resources that are required to run Amazon EKS pods on AWS Fargate.
- ☐ EKS - Fargate profile
Allows EKS to run Fargate tasks.
- ☐ EKS - Connector
Allows access to other AWS service resources that are required to connect to external clusters.
- ☐ EKS Local - Outpost
Allows Amazon EKS Local to call AWS services on your behalf.
- ☐ EKS - Pod Identity
Allows pods running in Amazon EKS cluster to access AWS resources.
- ☒ EKS - Auto Cluster
Allows access to other AWS service resources that are required to operate Auto Mode clusters managed by EKS.
- ☐ EKS - Auto Node
Allows EKS nodes to connect to EKS Auto Mode clusters and to pull container images from ECR.

Cancel

Next

Here, you can view the **permission policies**. After reviewing, click **Next**

Add permissions [Info](#)

Permissions policies (5) [Info](#)

The type of role that you selected requires the following policy.

Policy name ?	Type
AmazonEKSBLOCKStoragePolicy	AWS managed
AmazonEKSClusterPolicy	AWS managed
AmazonEKSComputePolicy	AWS managed
AmazonEKSLoadBalancingPolicy	AWS managed
AmazonEKSNetworkingPolicy	AWS managed

► Set permissions boundary - *optional*

Cancel

Previous

Next

Enter the **role name** and click **Create role**



Name, review, and create

Role details

Role name
Enter a meaningful name to identify this role.

Maximum 64 characters. Use alphanumeric and "+, -, @, _" characters.

Description
Add a short explanation for this role.

Maximum 1000 characters. Use letters (A-Z and a-z), numbers (0-9), tabs, new lines, or any of the following characters: ~, !, @, /, [] " ' =, < > .

Step 1: Select trusted entities Edit

Trust policy

```
1 {
2   "Version": "2002-10-17",
3   "Statement": [
4     {
5       "Effect": "Allow",
6       "Principal": {
7         "Service": "eks.amazonaws.com"
8       },
9       "Action": [
10        "sts:AssumeRole",
11        "sts:TagSession"
12      ]
13    }
14  ]
15 }
```

Step 2: Add permissions Edit

Permissions policy summary

Policy name	Type	Attached as
AmazonEKSBasedStoragePolicy	AWS managed	Permissions policy
AmazonEKSClusterPolicy	AWS managed	Permissions policy
AmazonEKSComputePolicy	AWS managed	Permissions policy
AmazonEKSElasticLoadBalancingPolicy	AWS managed	Permissions policy
AmazonEKSElasticNetworkPolicy	AWS managed	Permissions policy

Step 3: Add tags

Add tags - optional Info
Tags are key-value pairs that you can add to AWS resources to help identify, organize, or search for resources.
No tags associated with the resource.
[Add new tag](#)
You can add up to 50 more tags.

[Cancel](#) [Previous](#) [Create role](#)

The role has been successfully created for the **EKS Auto Cluster**

Role AmazonEKSAutoClusterRole created.

[View role](#)

Roles (14) [Info](#) [Refresh](#) [Delete](#) [Create role](#)

An IAM role is an identity you can create that has specific permissions with credentials that are valid for short durations. Roles can be assumed by entities that you trust.

☐

Role name

☐

[AmazonEKSAutoClusterRole](#)

▲

Trusted entities

AWS Service: eks

Last activity

-

Click **Create recommended role** to create the **Node IAM role**

Node IAM role [Info](#)
Nodes need an EC2 Instance IAM Role to launch and register with a cluster. To create a new custom role, follow the instructions in the [Amazon EKS User Guide](#).

[Refresh](#) [Create recommended role](#)

VPC [Info](#)
Select a VPC to use for your EKS cluster resources.

[Refresh](#) [Create VPC](#)

Subnets [Info](#)
Choose the subnets in your VPC where the control plane may place elastic network interfaces (ENIs) to facilitate communication with your cluster. To create a new subnet, go to the corresponding page in the [VPC console](#).

[Refresh](#) [Clear selected subnets](#)

subnet-0fbc55acae5948f0
eu-west-1a 172.31.32.0/20 Type: Public

subnet-0af35d04fa05f138f
eu-west-1b 172.31.0.0/20 Type: Public

subnet-069f3d23d47711158
eu-west-1c 172.31.16.0/20 Type: Public

Select **AWS Service** as the trusted entity



- Step 1
● **Select trusted entity**
- Step 2
○ Add permissions
- Step 3
○ Name, review, and create

Select trusted entity [Info](#)

Trusted entity type

☒ **AWS service**

Allow AWS services like EC2, Lambda, or others to perform actions in this account.

☐ **AWS account**

Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.

☐ **Web identity**

Allows users federated by the specified external web identity provider to assume this role to perform actions in this account.

☐ **SAML 2.0 federation**

Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.

☐ **Custom trust policy**

Create a custom trust policy to enable others to perform actions in this account.

Use case

Allow an AWS service like EC2, Lambda, or others to perform actions in this account.

Service or use case

EKS

Select **EKS** as the service name and **EKS Auto Node** as the use case



Use case

Allow an AWS service like EC2, Lambda, or others to perform actions in this account.

Service or use case

EKS

Choose a use case for the specified service.

Use case

- ☐ **EKS - Service**
Allows EKS to manage clusters on your behalf.
- ☐ **EKS - Cluster**
Allows the cluster Kubernetes control plane to manage AWS resources on your behalf.
- ☐ **EKS - Nodegroup**
Allow EKS to manage nodegroups on your behalf.
- ☐ **EKS - Fargate pod**
Allows access to other AWS service resources that are required to run Amazon EKS pods on AWS Fargate.
- ☐ **EKS - Fargate profile**
Allows EKS to run Fargate tasks.
- ☐ **EKS - Connector**
Allows access to other AWS service resources that are required to connect to external clusters
- ☐ **EKS Local - Outpost**
Allows Amazon EKS Local to call AWS services on your behalf.
- ☐ **EKS - Pod Identity**
Allows pods running in Amazon EKS cluster to access AWS resources.
- ☐ **EKS - Auto Cluster**
Allows access to other AWS service resources that are required to operate Auto Mode clusters managed by EKS.
- ☒ **EKS - Auto Node**
Allows EKS nodes to connect to EKS Auto Mode clusters and to pull container images from ECR.

Cancel

Next

Use case

Allow an AWS service like EC2, Lambda, or others to perform actions in this account.

Service or use case

EKS

Choose a use case for the specified service.

Use case

- ☐ **EKS - Service**
Allows EKS to manage clusters on your behalf.
- ☐ **EKS - Cluster**
Allows the cluster Kubernetes control plane to manage AWS resources on your behalf.
- ☐ **EKS - Nodegroup**
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Allows pods running in Amazon EKS cluster to access AWS resources.
- ☐ **EKS - Auto Cluster**
Allows access to other AWS service resources that are required to operate Auto Mode clusters managed by EKS.
- ☒ **EKS - Auto Node**
Allows EKS nodes to connect to EKS Auto Mode clusters and to pull container images from ECR.

Cancel

Next

The **permission policies** for **EKS Auto Node** are added. Click **Next** to proceed



Permissions policies (2) [Info](#)

Policy name	Type
 AmazonEC2ContainerRegistryPullOnly	AWS managed
AmazonEKSWorkerNodeMinimalPolicy	AWS managed

Cancel Previous Next

Name, review, and create

Cancel Previous Create role

 Delete
 Create role

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Cluster IAM role | [Info](#)
Select the Cluster IAM role to allow the Kubernetes control plane to manage AWS resources on your behalf. This cannot be changed after the cluster is created. To create a new custom role, follow the instructions in the [Amazon EKS User Guide](#).

AmazonEKSAutoClusterRole [Create recommended role](#)

Node IAM role | [Info](#)
Nodes need an EC2 Instance IAM Role to launch and register with a cluster. To create a new custom role, follow the instructions in the [Amazon EKS User Guide](#).

AmazonEKSAutoNodeRole [Create recommended role](#)

VPC | [Info](#)
Select a VPC to use for your EKS cluster resources.

vpc-01c128bb0e2b77712 | Default [Create VPC](#)

Subnets | [Info](#)
Choose the subnets in your VPC where the control plane may place elastic network interfaces (ENIs) to facilitate communication with your cluster. To create a new subnet, go to the corresponding page in the [VPC console](#).

Select subnets [Clear selected subnets](#)

subnet-0fbc55sacae5948f0
eu-west-1a 172.31.32.0/20 Type: Public

subnet-0af35d04fa05f138f
eu-west-1b 172.31.0.0/20 Type: Public

subnet-069f3d23d47711158
eu-west-1c 172.31.16.0/20 Type: Public

You can now view the **EKS Auto Mode capabilities**

▼ **View quick configuration defaults**

You can edit the following fields prior to creation by switching to "Custom configuration" in configuration options.

▼ Included capabilities with EKS Auto mode

EKS Auto Mode includes the following capabilities:

- Application load balancing
- Block Storage
- Compute Autoscaling
- GPU support
- Cluster DNS
- Pod and service networking

Here, you can see that the **EKS Auto Mode** value is set to **enabled**

Cluster configuration

Field	Value	Editable after the cluster is created
Cluster IAM role	AmazonEKSAutoClusterRole	No
EKS Auto Mode	Enabled View pricing	Yes
Node IAM role	AmazonEKSAutoNodeRole	No
Default node pools	general-purpose, system	Yes
Cluster administrator access	Allow cluster creator cluster administrator access	Yes
Cluster authentication mode	EKS API	No
Secrets encryption	Not enabled	Yes
Upgrade policy	Standard	Yes

Click **Create** to initialize the creation of the cluster



Networking

Field	Value	Editable after the cluster is created
VPC ID	vpc-01c128bb0e2b77712	No
Subnet IDs	subnet-0fbc55acae5948f0, subnet-0af35d04fa05f138f, subnet-069f3d23d47711158	Yes
Security groups	Create during cluster creation	Yes
Cluster endpoint access	Public and private	Yes
Cluster IP address family	IPv4	No
Public access source allowlist	0.0.0.0/0	Yes

Observability

Field	Value	Editable after the cluster is created
Cluster logging	api, audit, authenticator, controllerManager, scheduler	Yes

[Cancel](#)[Create](#)

The cluster creation request has been submitted, and it will take some time to complete.

Cluster creation request submitted
mahi-eks-auto-mode is being created. Learn how to [deploy a sample app](#). EKS will create nodes as needed.

[Delete cluster](#) [View dashboard](#)

Cluster info

Status
Creating

Kubernetes version
1.31

Support period
Standard support until November 26, 2025

Provider
EKS

Cluster health issues
0

Upgrade insights
0

Once the EKS cluster is created, you will see **EKS Auto Mode** displayed as **Enabled**

Amazon Elastic Kubernetes Service

Clusters

Amazon EKS Anywhere

Related services

Console settings

Documentation

Submit feedback

mahi-eks-auto-mode

[Delete cluster](#) [View dashboard](#)

Cluster info

Status
Active

Kubernetes version
1.31

Support period
Standard support until November 26, 2025

Provider
EKS

Cluster health issues
0

Upgrade insights
1

Overview

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Compute

Networking

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Access

Observability

Update history

Tags

Details

API server endpoint
<https://SEFA0A49C9AFAC3115D12E81D58768F4.y4.eu-west-1.eks.amazonaws.com>

Certificate authority
LS0hLS1CRUdJTiBDRVJUSUZJQ0FURSO0hS0ck1J5URCVENDQWUyZ0F3SUJBZ0UQREUvIrSWwzZkV3RFFZSktvWklodmNOQVFFTEJRQXdGVEVUJUJFR0E

OpenID Connect provider URL
<https://oidc.eks.eu-west-1.amazonaws.com/id/SEFA0A49C9AFAC3115D12E81D58768F4>

Cluster IAM role ARN
[arn:aws:iam::038462791702:role/AmazonEKSAutoClusterRole](#)
[View in IAM](#)

Created
2 hours ago

Cluster ARN
[arn:aws:eks:eu-west-1:038462791702:cluster/mahi-eks-auto-mode](#)

Platform version
eks.12

EKS Auto Mode

EKS automates routine cluster tasks for compute, storage, and networking to meet application compute needs.

EKS Auto Mode
Enabled

[Manage](#)



Step 3: Disable EKS Auto Mode

Go to the **EKS Cluster dashboard** and click on **Manager**

mahi-eks-auto-mode

[Delete cluster](#) [View dashboard](#)

Cluster info [Info](#)

Status
Active

Cluster health issues
0

Kubernetes version [Info](#)
1.31

Upgrade insights
1

Support period
Standard support until November 26, 2025

Provider
EKS

[Overview](#) [Resources](#) [Compute](#) [Networking](#) [Add-ons](#) [Access](#) [Observability](#) [Update history](#) [Tags](#)

Details

API server endpoint
<https://5EFA0A49C9AFAC3115D12E81D58768F4.yl4.eu-west-1.eks.amazonaws.com>

Certificate authority
LS0tLS1CRUdJTIBDRVJUSUZJQ0FUR50tLS0tCk1JSURCVENDQWUyZ0F3SUJBZ0UQ0REUViRSWwzZkV3RFFZSktvWklodmNOQVFFTEJRCQdGVGVUTUJFR0E=

OpenID Connect provider URL
<https://oidc.eks.eu-west-1.amazonaws.com/id/5EFA0A49C9AFAC3115D12E81D58768F4>

Cluster IAM role ARN
[arn:aws:iam::038462791702:role/AmazonEKSAutoClusterRole](#) [View in IAM](#)

Created
2 hours ago

Cluster ARN
[arn:aws:eks:eu-west-1:038462791702:cluster/mahi-eks-auto-mode](#)

Platform version [Info](#)
eks.12

EKS Auto Mode [Info](#)

EKS automates routine cluster tasks for compute, storage, and networking to meet application compute needs.

EKS Auto Mode
Enabled

[Manage](#)

Uncheck the **Use EKS Auto Mode** option to disable

[EKS](#) > [Clusters](#) > [mahi-eks-auto-mode](#) > Manage EKS Auto Mode

Manage EKS Auto Mode

EKS Auto Mode [Info](#)

EKS automates routine cluster tasks for compute, storage, and networking to meet application compute needs.

☒ **Use EKS Auto Mode**
EKS automates routine cluster tasks for compute, storage, and networking. When a new pod can't fit onto existing nodes, EKS creates a new node. EKS combines cluster infrastructure managed by AWS with integrated Kubernetes capabilities to meet application compute needs. [View pricing](#)

Included capabilities

Cluster IAM role
The cluster IAM role cannot be changed after the cluster is created, but the existing role can be updated. To update your cluster role, visit the IAM console.

Selected cluster IAM role
AmazonEKSAutoClusterRole

Built-in node pools - optional [Info](#)
EKS Auto Mode uses node pools to create nodes for pods. The node IAM role will be associated with built-in node pools. Use the Kubernetes API after cluster creation to create your own node pools.

Choose node pool(s)

general-purpose

system

Selected node IAM role
arn:aws:iam::038462791702:role/AmazonEKSAutoNodeRole

[Cancel](#)

[Save changes](#)

Click **Save changes**



EKS > Clusters > mahi-eks-auto-mode > Manage EKS Auto Mode

Manage EKS Auto Mode

EKS Auto Mode [Info](#)

EKS automates routine cluster tasks for compute, storage, and networking to meet application compute needs.

☒ Use EKS Auto Mode

EKS automates routine cluster tasks for compute, storage, and networking. When a new pod can't fit onto existing nodes, EKS creates a new node. EKS combines cluster infrastructure managed by AWS with integrated Kubernetes capabilities to meet application compute needs. [View pricing](#)

► Included capabilities

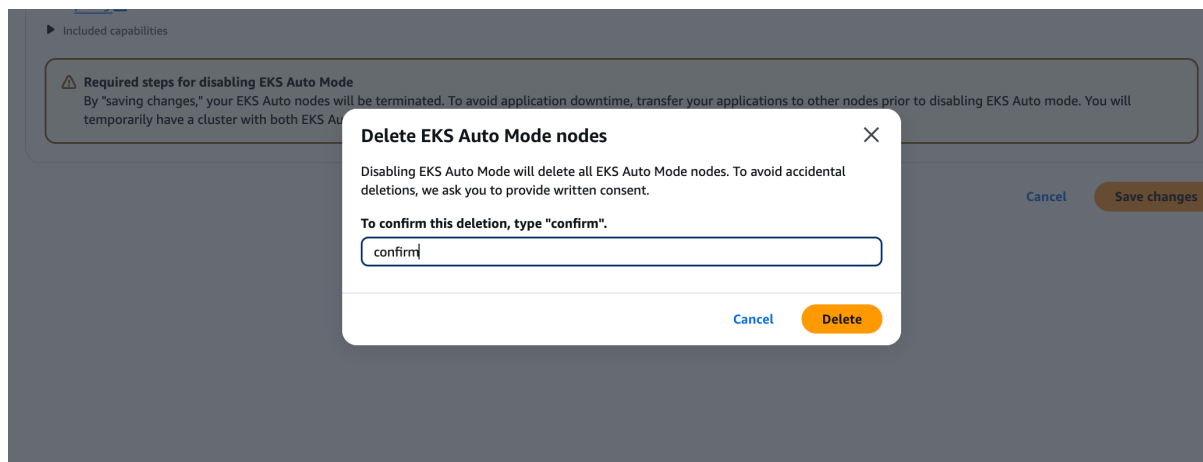
⚠ Required steps for disabling EKS Auto Mode

By "saving changes," your EKS Auto nodes will be terminated. To avoid application downtime, transfer your applications to other nodes prior to disabling EKS Auto mode. You will temporarily have a cluster with both EKS Auto Mode and other nodes until it is disabled.

[Cancel](#)

[Save changes](#)

Type **confirm** to delete the EKS Auto Mode nodes, then click **Delete**



Now, you will see the EKS Auto Mode status updated to Disabled

mahi-eks-auto-mode

[Refresh](#)[Delete cluster](#)[View dashboard](#)

▼ Cluster info [Info](#)

Status

✔ Active

Cluster health issues

✔ 0

Kubernetes version [Info](#)

1.31

Upgrade insights

✔ 1

Support period

📅 Standard support until November 26, 2025

Provider

EKS

Overview

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API server endpoint

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OpenID Connect provider URL

🔗 <https://oidc.eks.eu-west-1.amazonaws.com/id/5EFA0A49C9AFAC3115D12E81D58768F4>

Cluster IAM role ARN

🔗 <arn:aws:iam::038462791702:role/AmazonEKSAutoClusterRole>
[View in IAM](#)

Created

🕒 2 hours ago

Cluster ARN

🔗 <arn:aws:eks:eu-west-1:038462791702:cluster/mahi-eks-auto-mode>

Platform version [Info](#)

eks.12

EKS Auto Mode [Info](#)

EKS automates routine cluster tasks for compute, storage, and networking to meet application compute needs.

EKS Auto Mode

Disabled

[Manage](#)



Method 2: Create an EKS Auto Mode Cluster with eksctl

To create an EKS Auto Mode cluster using eksctl, ensure you have the eksctl CLI installed. Then, run this command: **eksctl create cluster --name=devops-auto-mode --enable-auto-mode**. This will create a cluster named devops-auto-mode with EKS Auto Mode enabled

```
mahendransaselvakumar@Mahendransas-MBP ~ % eksctl create cluster --name=devops-auto-mode --enable-auto-mode
Error: unknown flag: --enable-auto-mode
```

If you encounter the error "**unknown flag: --enable-auto-mode**", it means your eksctl version is outdated.

Upgrade the eksctl version to the latest release

Run the **brew update** command to update Homebrew

```
mahendransaselvakumar@Mahendransas-MBP ~ % brew update
==> Updating Homebrew...
Updated 4 taps (hashicorp/tap, weaveworks/tap, homebrew/core and homebrew/cask).
==> New Formulae
aliase                azure-core-cpp        eslint_d              libunicond            rshijack
ampl-asl              cargo-chef            eventpp               libxsimd-frontent     scooter
ansible@10           cargo-expand          fltk@1.3             lla                   sesh
```

Run the **brew upgrade eksctl** command to upgrade eksctl to the latest version

```
mahendransaselvakumar@Mahendransas-MBP ~ % brew upgrade eksctl
==> Upgrading 1 outdated package:
weaveworks/tap/eksctl 0.190.0 -> 0.196.0
==> Fetching dependencies for weaveworks/tap/eksctl: aws-iam-authenticator and kubernetes-cli
==> Fetching aws-iam-authenticator
==> Downloading https://ghcr.io/v2/homebrew/core/aws-iam-authenticator/manifests/0.6.28
##### 100.0%
==> Downloading https://ghcr.io/v2/homebrew/core/aws-iam-authenticator/blobs/sha256:2247afcae2b6628195756935701bc46d873d98ae8
##### 100.0%
==> Fetching kubernetes-cli
==> Downloading https://ghcr.io/v2/homebrew/core/kubernetes-cli/manifests/1.31.3
##### 100.0%
==> Downloading https://ghcr.io/v2/homebrew/core/kubernetes-cli/blobs/sha256:118193fcf9a4f91c21f0583589388ae6dc287963e591642f
##### 100.0%
==> Fetching weaveworks/tap/eksctl
==> Downloading https://github.com/eksctl-io/eksctl/releases/download/v0.196.0/eksctl_Darwin_arm64.tar.gz
==> Downloading from https://objects.githubusercontent.com/github-production-release-asset-2e65be/134539560/9e30a5bb-0b14-45c
##### 100.0%
==> Upgrading weaveworks/tap/eksctl
0.190.0 -> 0.196.0
==> Installing dependencies for weaveworks/tap/eksctl: aws-iam-authenticator and kubernetes-cli
==> Installing weaveworks/tap/eksctl dependency: aws-iam-authenticator
==> Downloading https://ghcr.io/v2/homebrew/core/aws-iam-authenticator/manifests/0.6.28
Already downloaded: /Users/mahendransaselvakumar/Library/Caches/Homebrew/downloads/d17a9e218d5379106bf572bd9994bd2cca2ed29327277aeddbaf351236d643--aws-iam-authenticat
or-0.6.28.bottle.manifest.json
==> Pouring aws-iam-authenticator--0.6.28.arm64_sonoma.bottle.tar.gz
📦 /opt/homebrew/Cellar/aws-iam-authenticator/0.6.28: 7 files, 59.1MB
==> Installing weaveworks/tap/eksctl dependency: kubernetes-cli
==> Downloading https://ghcr.io/v2/homebrew/core/kubernetes-cli/manifests/1.31.3
Already downloaded: /Users/mahendransaselvakumar/Library/Caches/Homebrew/downloads/f8fd19d10e239038f339af3c9b47978cb154932f089fbf6b7d67ea223df378de--kubernetes-cli-1.31
.3.bottle.manifest.json
==> Pouring kubernetes-cli--1.31.3.arm64_sonoma.bottle.tar.gz
📦 /opt/homebrew/Cellar/kubernetes-cli/1.31.3: 237 files, 60.2MB
==> Installing weaveworks/tap/eksctl
==> Caveats
zsh completions have been installed to:
/opt/homebrew/share/zsh/site-functions
==> Summary
📦 /opt/homebrew/Cellar/eksctl/0.196.0: 6 files, 155.3MB, built in 3 seconds
==> Running 'brew cleanup eksctl' ...
Disable this behaviour by setting HOMEBREW_NO_INSTALL_CLEANUP.
Hide these hints with HOMEBREW_NO_ENV_HINTS (see 'man brew').
Removing: /opt/homebrew/Cellar/eksctl/0.190.0... (6 files, 151MB)
==> Caveats
```



Now, run the command **eksctl create cluster --name=devops-auto-mode --enable-auto-mode** to create the EKS Auto Mode cluster

```
700mahendranselvakumar@Mahendrans-MBP ~ % eksctl create cluster --name=devops-auto-mode --enable-auto-mode
2024-12-02 23:02:06 [i] eksctl version 0.196.0
2024-12-02 23:02:06 [i] using region eu-west-1
2024-12-02 23:02:06 [i] setting availability zones to [eu-west-1a eu-west-1c eu-west-1b]
2024-12-02 23:02:06 [i] subnets for eu-west-1a - public:192.168.0.0/19 private:192.168.96.0/19
2024-12-02 23:02:06 [i] subnets for eu-west-1c - public:192.168.32.0/19 private:192.168.128.0/19
2024-12-02 23:02:06 [i] subnets for eu-west-1b - public:192.168.64.0/19 private:192.168.160.0/19
2024-12-02 23:02:06 [i] using Kubernetes version 1.30
2024-12-02 23:02:06 [i] creating EKS cluster "devops-auto-mode" in "eu-west-1" region with
2024-12-02 23:02:06 [i] if you encounter any issues, check CloudFormation console or try 'eksctl utils describe-stacks --region=eu-west-1 --cluster=devops-auto-mode'
2024-12-02 23:02:06 [i] Kubernetes API endpoint access will use default of {publicAccess=true, privateAccess=false} for cluster "devops-auto-mode" in "eu-west-1"
2024-12-02 23:02:06 [i] CloudWatch logging will not be enabled for cluster "devops-auto-mode" in "eu-west-1"
2024-12-02 23:02:06 [i] you can enable it with 'eksctl utils update-cluster-logging --enable-types={SPECIFY-YOUR-LOG-TYPES-HERE (e.g. all)} --region=eu-west-1 --cluster=devops-auto-mode'
2024-12-02 23:02:06 [i]
2 sequential tasks: { create cluster control plane "devops-auto-mode", wait for control plane to become ready
}
2024-12-02 23:02:06 [i] building cluster stack "eksctl-devops-auto-mode-cluster"
2024-12-02 23:02:06 [i] deploying stack "eksctl-devops-auto-mode-cluster"
2024-12-02 23:02:36 [i] waiting for CloudFormation stack "eksctl-devops-auto-mode-cluster"
2024-12-02 23:03:06 [i] waiting for CloudFormation stack "eksctl-devops-auto-mode-cluster"
2024-12-02 23:04:07 [i] waiting for CloudFormation stack "eksctl-devops-auto-mode-cluster"
2024-12-02 23:05:07 [i] waiting for CloudFormation stack "eksctl-devops-auto-mode-cluster"
2024-12-02 23:06:07 [i] waiting for CloudFormation stack "eksctl-devops-auto-mode-cluster"
2024-12-02 23:07:07 [i] waiting for CloudFormation stack "eksctl-devops-auto-mode-cluster"
2024-12-02 23:08:07 [i] waiting for CloudFormation stack "eksctl-devops-auto-mode-cluster"
2024-12-02 23:09:07 [i] waiting for CloudFormation stack "eksctl-devops-auto-mode-cluster"
2024-12-02 23:10:08 [i] waiting for CloudFormation stack "eksctl-devops-auto-mode-cluster"
2024-12-02 23:11:08 [i] waiting for CloudFormation stack "eksctl-devops-auto-mode-cluster"
2024-12-02 23:12:08 [i] waiting for CloudFormation stack "eksctl-devops-auto-mode-cluster"
2024-12-02 23:13:08 [i] waiting for CloudFormation stack "eksctl-devops-auto-mode-cluster"
2024-12-02 23:15:09 [i] waiting for the control plane to become ready
2024-12-02 23:15:10 [✓] saved kubeconfig as "/Users/mahendranselvakumar/.kube/config"
2024-12-02 23:15:10 [i] no tasks
2024-12-02 23:15:10 [✓] all EKS cluster resources for "devops-auto-mode" have been created
2024-12-02 23:15:11 [i] kubectl command should work with "/Users/mahendranselvakumar/.kube/config", try 'kubectl get nodes'
2024-12-02 23:15:11 [✓] EKS cluster "devops-auto-mode" in "eu-west-1" region is ready
mahendranselvakumar@Mahendrans-MBP ~ %
```

You will be able to view the newly created EKS cluster in the AWS Management Console

Clusters (2) Info								
Filter clusters			Delete Create cluster		< 1 >			
	Cluster name	Status	Kubernetes version	Support period	Upgrade policy	Created	Provider	
☒	devops-auto-mode	Active	1.30 Upgrade now	Standard support until July 28, 2025	Extended	20 minutes ago	EKS	
☐	mahi-eks-auto-mode	Active	1.31	Standard support until November 26, 2025	Standard	3 hours ago	EKS	

Navigate to the newly created EKS cluster in the AWS Management Console, where you will see that **EKS Auto Mode** is set to **Enabled**



devops-auto-mode

[Delete cluster](#)[Upgrade version](#)[View dashboard](#)

ⓘ Your current IAM principal doesn't have access to Kubernetes objects on this cluster. This may be due to the current user or role not having Kubernetes RBAC permissions to describe cluster resources or not having an entry in the cluster's auth config map. [Learn more](#)

ⓘ End of standard support for Kubernetes version 1.30 is July 28, 2025. On that date, your cluster will enter the extended support period with additional fees. For more information, see the [pricing page](#). [Upgrade now](#)

▼ Cluster info [Info](#)

Status
Active

Kubernetes version [Info](#)
1.30

Support period
Standard support until July 28, 2025

Provider
EKS

Cluster health issues

0

Upgrade insights

0

Overview

[Resources](#)[Compute](#)[Networking](#)[Add-ons](#)[Access](#)[Observability](#)[Update history](#)[Tags](#)

Details

API server endpoint

<https://99FBCF0CA95DC82AE88FC34F8D6D3A9.gr7.eu-west-1.eks.amazonaws.com>

OpenID Connect provider URL

<https://oidc.eks.eu-west-1.amazonaws.com/id/99FBCF0CA95DC82AE88FC34F8D6D3A9>

Created

21 minutes ago

Certificate authority

[LS0HLS1CRUJDTIBDRVJUSUZJQ0FURSOHL5QtcK1JSURCVENDQWUyZ0F3SUJBZ0JVVFFHYVRxd09NYkl3RFFZSkVWklodmNQQVFFTEJRQXdGVEVUTUJFR0ExVUUKQXhNS2EzVmtaWEp1](#)

Cluster IAM role ARN

[arn:aws:iam::038462791702:role/eksctl-devops-auto-mode-cluster-ServiceRole-LVNRBGs7kZgA](#)

[View in IAM](#)

Cluster ARN

[arn:aws:eks:eu-west-1:038462791702:cluster/devops-auto-mode](#)

Platform version [Info](#)
eks.20

EKS Auto Mode [Info](#)

EKS automates routine cluster tasks for compute, storage, and networking to meet application compute needs.

EKS Auto Mode
Enabled

[Manage](#)

Disabling **EKS Auto Mode** is not possible using the `eksctl` command. To remove Auto Mode, you need to delete the existing cluster and recreate it without enabling Auto Mode. To delete the cluster, run this command **`eksctl delete cluster --name=devops-auto-mode`**

```
mahendransaselvakumar@Mahendransas-MBP ~ % eksctl delete cluster --name=devops-auto-mode
2024-12-02 23:27:54 [i] deleting EKS cluster "devops-auto-mode"
2024-12-02 23:27:54 [i] deleted 0 Fargate profile(s)
2024-12-02 23:27:54 [✓] kubeconfig has been updated
2024-12-02 23:27:54 [i] cleaning up AWS load balancers created by Kubernetes objects of Kind Service or Ingress
2024-12-02 23:27:55 [i] 1 task: { delete cluster control plane "devops-auto-mode" [async] }
2024-12-02 23:27:55 [i] will delete stack "eksctl-devops-auto-mode-cluster"
2024-12-02 23:27:56 [✓] all cluster resources were deleted
mahendransaselvakumar@Mahendransas-MBP ~ %
```

Conclusion

EKS Auto Mode simplifies Kubernetes cluster management by automating infrastructure and operational tasks, making it easier to deploy and maintain production-ready environments. With features like automated scaling, cost optimization, enhanced security, and seamless AWS integration, it provides a balanced approach to managing workloads effectively while reducing operational overhead.

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Keep Learning, Keep EKS-ing!!

Feel free to reach out to me, if you have any other queries or suggestions Stay connected on LinkedIn: [Mahendran Selvakumar](#)

Stay connected on Medium: <https://devopstronaut.com/>